

Financial Sample Case Study

Explanation Report

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Dataset: Financial Sample (Microsoft), 700 orders

This document walks through how I approached the case study — cleaning the data, analyzing it with SQL and Pandas, and building out visualizations in Python, Excel, and Tableau. It's meant to explain the reasoning behind each step, not just list the outputs, so it doubles as the demo/explanation deliverable alongside the code itself.

What's in the code folder

File / folder	What it does
01_clean_data.py	Loads the raw Excel file, checks it, fixes the missing Discount Band values
02_sql_analysis.py	Loads the cleaned data into SQLite and runs the analysis queries
03_visualize.py	Builds the 6 Matplotlib/Seaborn charts
build_excel_dashboard.py	Generates the Excel workbook with formulas + charts + dashboard tab
End_to_End_Analysis.ipynb	Same pipeline, one notebook, already run so the output is visible
data/	Raw file, cleaned CSV, and the SQLite database
charts/	The 6 PNGs from the Python step
excel_dashboard/	Financial_Analysis_Dashboard.xlsx
bi_tableau/	Tableau dashboard screenshot + notes on how it's built
End_to_End_Analysis_Report.pdf	The final written report — findings and recommendations

1. Cleaning the data

I started with the raw Financial Sample file — 700 rows, 16 columns. First thing I checked was duplicates (none) and missing values. There were 53 blank rows in Discount Band, so I looked closer and every single one of them had a Discounts value of 0. That made the fix obvious: those rows just weren't discounted at all, so I labeled them 'None' instead of dropping them or trying to guess a value.

I also wanted to make sure the calculated columns actually add up, since a lot of these public sample datasets have small inconsistencies baked in. I checked all three:

Field	Should equal	Mismatches
Gross Sales	Units Sold x Sale Price	0
Sales	Gross Sales - Discounts	0
Profit	Sales - COGS	0

Zero mismatches across all three, which was a nice surprise — it meant I didn't need to touch the numbers themselves, just the one categorical field. The cleaned file gets saved to `data/cleaned_financial_data.csv` and that's the file every other step in this project reads from, so everything downstream (SQL, Excel, Tableau) is working off the same source.

2. Analyzing it with SQL and Pandas

Once the data was clean, I loaded it into a small SQLite database (`data/financial_data.db`) so I could write real SQL against it instead of just doing everything in Pandas. Partly this was the brief, but it's also just easier to sanity-check a GROUP BY in SQL than to trust a chain of Pandas operations blindly.

Profit by segment

This is the finding that everything else in the report circles back to. Four of the five segments are solidly profitable. Enterprise is not:

Segment	Sales	Profit	Margin
Government	\$52.50M	\$11.39M	21.7%
Small Business	\$42.43M	\$4.14M	9.8%
Channel Partners	\$1.80M	\$1.32M	73.1%
Midmarket	\$2.38M	\$0.66M	27.7%
Enterprise	\$19.61M	-\$0.61M	-3.1%

What stood out to me is that Enterprise isn't a small, ignorable segment — it's the second-largest by sales. So this isn't a rounding error, it's a real problem worth digging into.

Does discounting explain it?

Grouping by Discount Band, margin drops in a pretty clean line as the discount gets deeper: 21.9% with no discount, down to 17.9%, then 14.4%, then just 9.1% at the High band. That's a pattern, but it doesn't explain why one specific segment goes negative while the others stay healthy even at High discount.

Segment x Discount Band

Crossing the two together answers that. Every other segment stays profitable even at the High discount band. Enterprise doesn't — it goes negative starting around the Medium band and gets worse from there. Of the 58 orders that lost money in the whole dataset, 33 were Enterprise deals in the High discount band. So it's not discounting in general that's the problem, it's specifically how Enterprise deals get discounted.

Products and countries

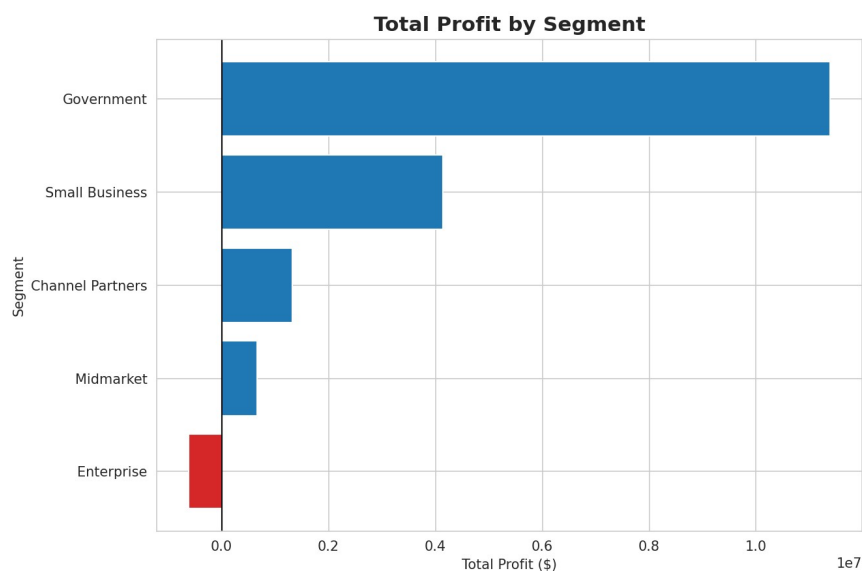
Paseo brings in the most revenue by a wide margin (\$33.0M), but it's not the most efficient — that's Amarilla, at close to 16% margin. Geography turned out to be a non-issue: all five countries sit in a tight 12-16% margin band, so wherever the losses are coming from, it isn't location.

Seasonality

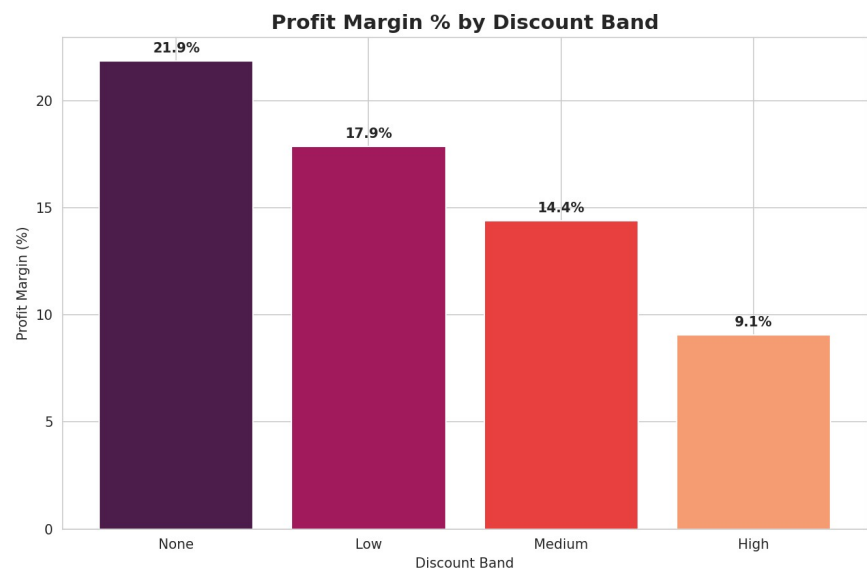
Sales and profit move together almost every month, which makes sense, but the two standout months every year are October and December. Worth keeping in mind for planning inventory or staffing around those periods.

3. Charting it in Python

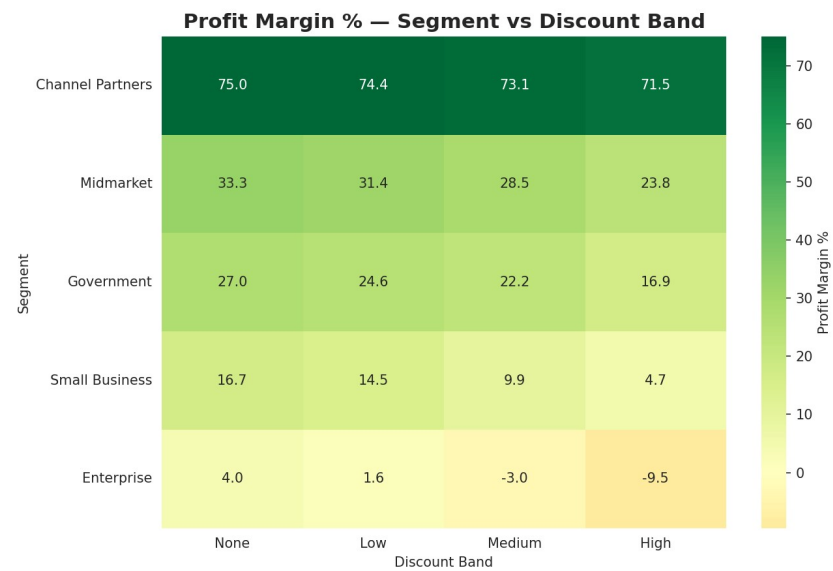
03_visualize.py builds these with Matplotlib and Seaborn, reading straight from the cleaned CSV. Same six charts that appear in the final written report.



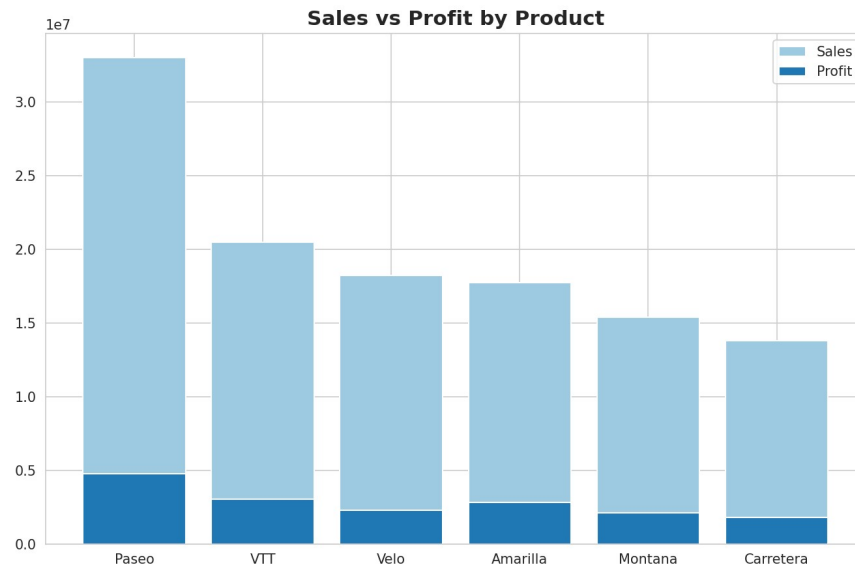
Profit by segment — Enterprise is the only bar in the red.



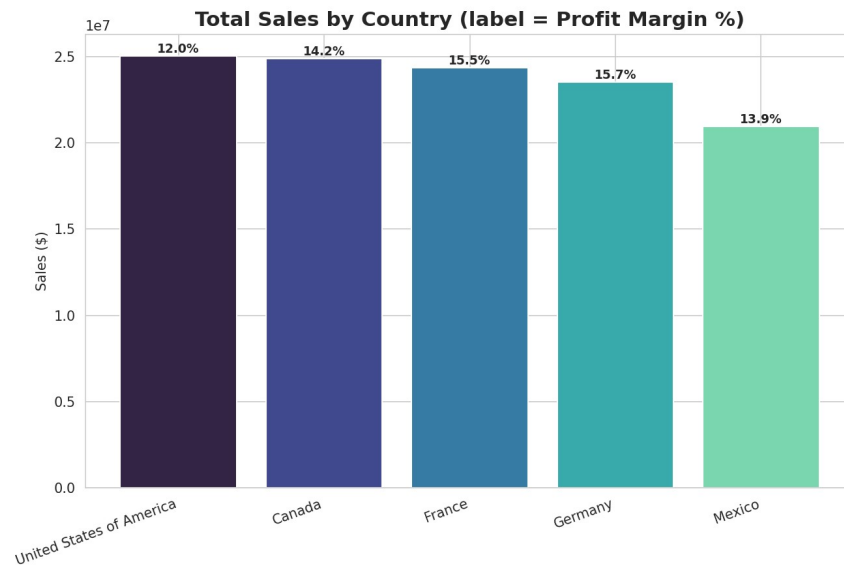
Margin drops steadily as discount depth increases.



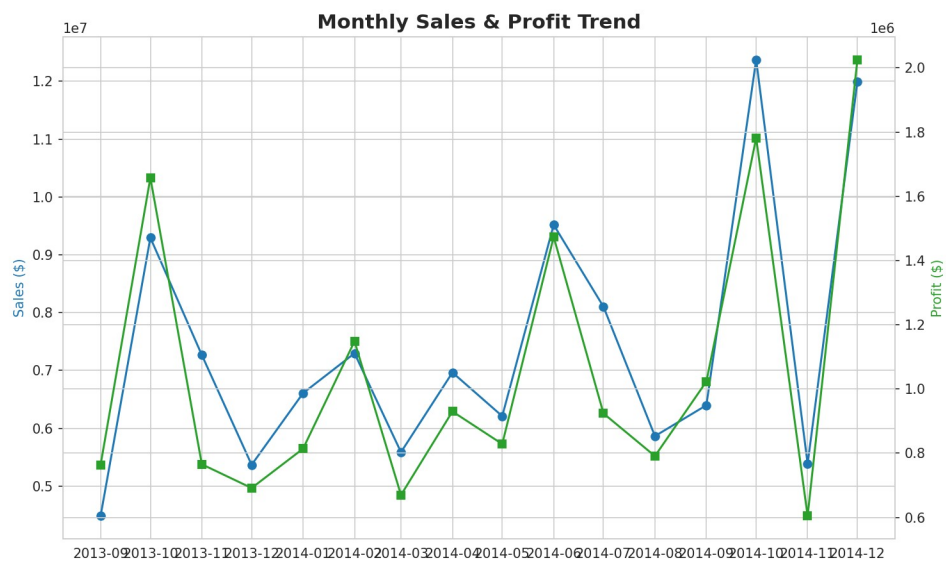
This is the one that made the Enterprise problem click for me — the red cells are all Enterprise, all Medium/High discount.



Sales vs profit by product.



Sales by country, labeled with margin — nothing unusual here.



Sales and profit over time. October/December spikes are visible in both years.

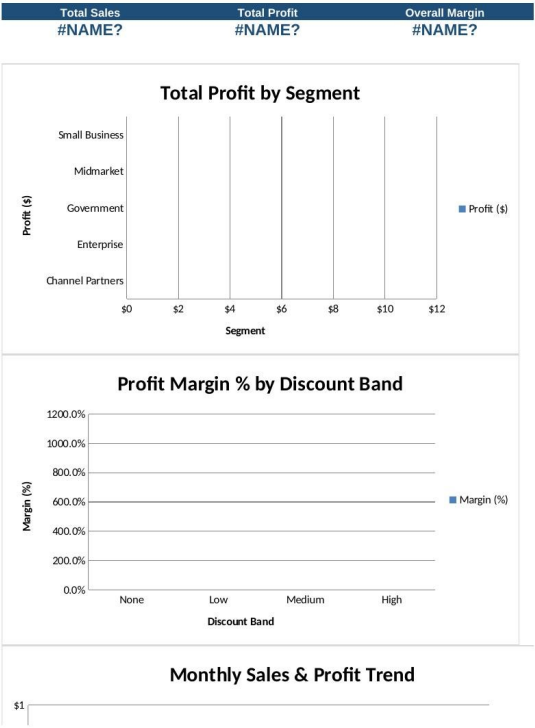
4. Excel dashboard

The brief asked for Excel specifically, and I didn't want to just paste static numbers in — I wanted the workbook to actually recalculate if the underlying data changed, so it's built with SUMIF/SUMIFS formulas pulling from a Raw Data tab of all 700 rows, not hardcoded totals.

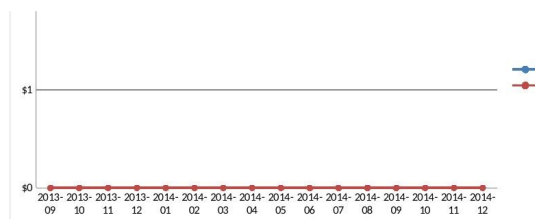
There's a summary tab for each of the analyses above (segment, discount band, the segment/discount cross-tab as a conditional-formatting heatmap, product, country, monthly trend), and a Dashboard tab up front that pulls KPI tiles and copies of the charts together. I ran it through a formula recalculation check afterward and all 116 formulas came back clean, no errors.

Financial Sample — Executive Dashboard

Sales, Discounts & Profitability — 700 orders, 5 segments, 6 products, 5 countries, Sep 2013–Dec 2014



Dashboard tab, top half — KPI tiles and the segment/discount-band charts.

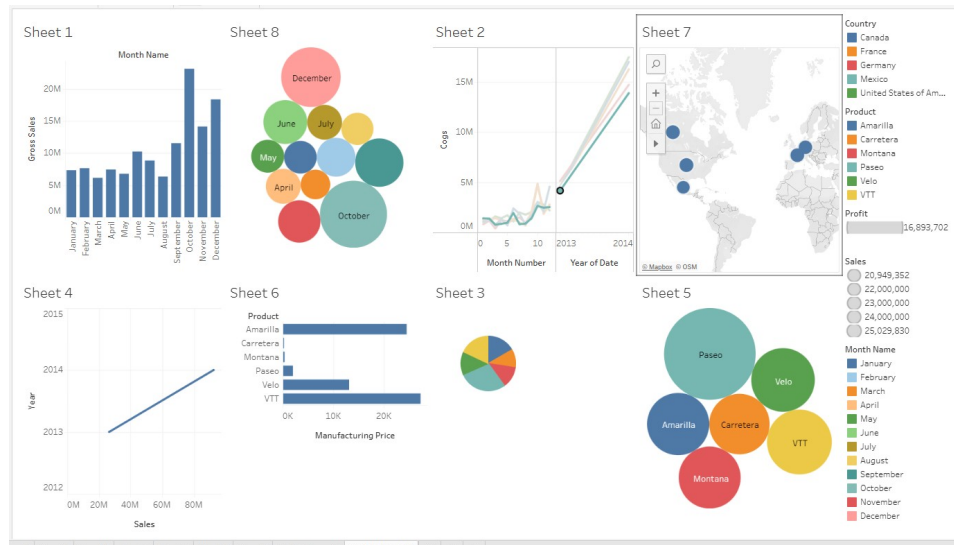


Dashboard tab, continued — product and country charts.

One thing worth flagging: the Total Profit figure here comes out to \$16,893,702, which is the same number the Tableau dashboard shows in the next section. That wasn't planned, it's just what happens when both tools are reading from the same cleaned data — but it's a decent gut-check that nothing got mangled along the way.

5. Tableau dashboard

This is the BI-tool half of the visualization requirement. I built it in Tableau, connected straight to the cleaned CSV, and put together 8 sheets on one dashboard with shared filters (Country, Product, Sales, Profit, Month Name) so you can slice everything at once.



The full dashboard as built in Tableau Desktop.

Sheet	Type	Shows
Sheet 1	Bar chart	Gross Sales by month — Oct/Dec stand out
Sheet 2	Line chart	COGS trend by month, split by year
Sheet 3	Pie chart	Sales mix
Sheet 4	Line chart	Sales by year
Sheet 5	Bubbles	Sales by product
Sheet 6	Bar chart	Manufacturing price by product
Sheet 7	Map	Profit by country
Sheet 8	Bubbles	Gross sales by month

If a native .twbx file is needed rather than just the screenshot, [bi_tableau/README.md](#) has the exact steps to rebuild each sheet — it's straightforward since every field it uses already exists in the cleaned CSV, no calculated fields required.

6. What I found

- Enterprise is the only segment losing money overall (-3.13% margin), and it's concentrated almost entirely in High-discount deals.
- Channel Partners is quietly the most efficient segment in the business — 73.1% margin, even though it's the smallest by revenue.
- Margin and discount depth move in lockstep for everyone except Enterprise, where it falls off a cliff instead of gradually.
- Paseo sells the most, Amarilla is the most profitable per dollar of sales.
- Country doesn't matter much here — all five sit in a healthy, similar margin range.

- October and December are reliably the strongest months, both years.

7. Recommendations

Based on the above, here's what I'd actually do with this:

- Put a hard cap on discounting for Enterprise deals, especially in the High band — this is where essentially all of the company's losses are coming from.
- Figure out why Enterprise needs such deep discounts in the first place. My guess is competitive pressure or long sales cycles, but that needs someone closer to the sales process to confirm. Either way, bundling or value-based pricing is probably a better lever than just cutting price further.
- Channel Partners' margin is high enough that it's worth asking whether that motion can be scaled up rather than just left as-is.
- Amarilla and VTT hold their margin well even when discounted, so they're good anchors for bundles or upsells.
- Line up inventory, staffing, and marketing spend with the October-December peak rather than spreading it evenly.

8. Running it yourself

From inside the project folder:

```
pip install -r requirements.txt
python 01_clean_data.py
python 02_sql_analysis.py
python 03_visualize.py
python build_excel_dashboard.py
```

Or just open End_to_End_Analysis.ipynb in Jupyter — it's already been run once so you can read the output without re-executing anything, but Kernel > Restart & Run All will reproduce it from scratch. For Tableau, point Tableau Desktop at data/cleaned_financial_data.csv and follow bi_tableau/README.md.

A quick way to sanity-check that everything ran correctly: the Enterprise margin should come out to -3.13% and Total Profit should land at \$16,893,702, in the notebook, the Excel dashboard, and the Tableau KPI alike. If those three numbers agree, the pipeline ran the way it's supposed to.